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Commodity price co-movement: heterogeneity and the time-varying impact of fundamentals

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Abstract

This paper extends the topical literature on the co-movement and determinants of primary commodity prices, by considering heterogeneity in commodities and time variation in the impact of fundamentals. We account for heterogeneity by employing a dynamic hierarchical factor model, which decomposes commodities into global and sectoral factors. Using a time-varying parameter factor augmented VAR model, we shock global and sector-specific factors over time. We present plausible impulse responses to demand shocks, real interest rate shocks and to elevated risks during the global financial crisis. We also identify that agricultural raw materials, food and metals respond heterogeneously to these shocks.

Keywords: commodity prices, co-movement, dynamic hierarchical factor models, time-varying parameter factor augmented VAR models

JEL classification: E3, F3, F4, G1

1. Introduction

Synchronised surges and declines in primary commodity prices have been the catalyst for a lively recent debate about their commonalities and determinants. These studies document a significant degree of co-movement in commodity prices, which can be modelled by a common factor. See recent commodity prices research by Cuddington and Jerrett (2008); Vansteenkiste (2009), Byrne, Fazio and Fiess (2013), Alquist and Coibion (2014), West and Wong (2014), Daskalaki, Kostakis and Skiadopoulos (2014), Yin and Han (2015), Antonakakis and Kizys (2015) and Alam and Gilbert (2017). A potential criticism of factor models with a single common element, however, is that they can forsake useful information. If commodity prices display

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important heterogeneity and/or are impacted by different fundamentals, then a single factor extracted from all commodities may not fully reflect price dynamics (Moench, Ng and Potter, 2013). We contribute therefore to the commodities literature by decomposing commodity prices into common and group factors using a hierarchical factor model and examine commodities' relationships with fundamentals.

To the extent that commodity prices share common determinants, the literature is ambiguous as to what drives recent movements, possibly because heterogeneity is unaccounted for, or the relative importance of determinants change over time. Among possible determinants, Wolf (2008), Svensson (2008), Frankel (2014) and Ratti and Vespignani (2015) have underlined that shifts in global demand matter for commodity prices. Interest rates have been emphasised by Frankel (2008, 2014) and Svensson (2008). Also, Beck (1993, 2001) indicated that uncertainty is an important determinant of primary commodity prices. A limited number of empirical studies have so far tested these different hypotheses and they rely on a time-invariant methodology (e.g. Byrne, Fazio and Fiess, 2013; Poncela, Senra and Sierra, 2014; Ratti and Vespignani, 2015; Alam and Gilbert, 2017). Hence, previous work on the determinants of commodity prices implicitly assumed that the impact of macroeconomic shocks on commodity prices does not vary over time and we investigate this assumption.

There are several reasons to believe that the relationship between primary commodity prices and macroeconomics fundamentals may be unstable (Alvarez-Ramirez *et al.*, 2012). For example, China has significantly increased its market shares of global commodities following its rapid development and this may impact demand effects (e.g. Kilian, 2009; Roache, 2012). Financial investors' risk-bearing appetite and risk premium may vary over time (Cheng and Xiong, 2014). Another potential cause of time-varying commodity effect is due to variation in commodity market participants. Since the 2000s, there has been a large inflow of investment capital from speculators, which has added to commodity market activity from commercial hedgers (e.g. farmers, producers and consumers) and non-commercial traders such as financial institutions (e.g. Cheng and Xiong, 2014).[1]

This paper therefore empirically examines the relationship between commodity common factors and fundamentals, while accounting for heterogeneity and potential instability. Our analysis incorporates important innovations relative to previous studies. First, we fully account for the determinants of commodity prices while allowing for price heterogeneity using a dynamic hierarchical factor model (DHFM). The existing literature typically explored the impact of macro fundamentals on a single aggregate common factor. But each sector may have a heterogeneous market structure, level of competition and concentration and a differing elasticity of demand (e.g. Yin and Han,

1 Commodity markets facilitate risk sharing among a broad set of agents, institutional investors have recently increased their portfolio allocations to commodities (Daskalaki, Kostakis and Skiadopoulos, 2014).

2015). The DHFM allows us to assess whether agricultural raw material, food and metal sectors respond differently to macroeconomics shocks. Importantly, the DHFM distinguishes commodity common and sector heterogeneous structures, and therefore it differs from the approach that extracts commodity factors from each commodity sector.

Our second contribution is to employ a time-varying parameter factor augmented vector autoregression (TVP-FAVAR) model with stochastic volatility to flexibly delineate the short-run impact of fundamentals. This approach allows all parameters to evolve continuously, informing us when, and to what extent, changes have occurred over time; rather than imposing an arbitrary sample split to account for changing dynamics. Our model also allows for time-varying heteroskedasticity in the VAR innovations to account for changes in the magnitude of shocks. This feature is especially important given intense commodity price and macroeconomic volatility between the Great Moderation and Global Financial Crisis (Primiceri, 2005; Baumeister and Peersman, 2013).

Our findings can be summarised as follows. We discover an important degree of commodity price co-movement due to common and sectoral factors, highlighting the importance of commodity heterogeneity. Commodities can be modelled therefore by a single common factor, but there are notable differences between commodities, especially during episodes of extreme price volatility. Next, we report and discuss the estimation results from time-invariant and time-varying FAVAR models. Under both approaches, we provide empirical evidence that macro fundamentals affect the returns of a large number of commodities. The impulse responses indicate qualitative and quantitative changes over time in commodity prices to demand, real interest rate and uncertainty shocks. We provide additional evidence that fundamental shocks relate differently to the material, food and metal sectors. For example, demand shocks have a more powerful impact on the metals' sector than others, while materials are more sensitive to uncertainty. Intuitively, and supporting the use of a time-varying methodology, we also find Chinese demand shocks have had an increasingly positive impact on commodity prices.

The rest of the paper is organised as follows. Section 2 reviews the related literature. Section 3 formally presents our econometric methodology. Section 4 discusses the data. Section 5 reports the empirical results and robustness checks. Section 6 offers some concluding remarks.

2. Brief literature review

Changes in commodity prices can be rapid, and large price swings severely impact commodity importers, exporters and speculators.[2] Thus, a better understanding of the nature of commodity prices and their determinants may

2 For example, higher commodity prices may lead to lower aggregate demand and production outputs, induce inflationary tendencies and higher interest rates for importing countries; whereas a sustained decline in commodity prices supports the so-called 'resource curse' hypothesis for commodity abundant emerging economies. See, among the others, Frankel

lead to better decision making in areas such as macroeconomic policy, risk and portfolio management. A long-standing literature focused on commodity prices' time series properties (e.g. Cuddington, 1992; Deaton, 1999; Cashin, Liang and McDermott, 2000). This is related to trends in primary commodity prices relative to the price of manufactured goods, within the context of the Prebisch (1950) and Singer (1950) hypothesis.[3] The literature has also considered commonalities in commodity prices. The seminal work by Pindyck and Rotemberg (1990), for example, was the first to confirm that prices of seemingly unrelated commodities tend to co-move. Deaton (1999) stressed that it is important to consider the time series properties of individual commodities and their co-movement, to assess the different impact of commodity prices on developing and industrial countries, and therefore the need for stabilisation policies. Cashin, McDermott and Scott (2002) also found evidence of price synchronisation of related commodities.

The co-movement of commodity prices since the turn of the 21st century has promoted a renewed interest in commonalities and sought to explain why prices co-move. Alquist and Coibion (2014) employed a general equilibrium model to decompose the sources of commodity price co-movement. They evidenced indirect shocks that impact on commodity prices through the change in aggregate output are main sources for commodity price commonalities. This empirical literature has, for example, employed factor models to extract commonalities. Cuddington and Jerrett (2008) used principal component analysis to investigate the degree of concordance between metal commodity prices. Panel time series methods were utilised by Byrne, Fazio and Fiess (2013) and they found evidence of co-movement of a large number of commodities due to one common factor. Chen *et al.* (2014) showed that the movements of 51 tradeable commodities were mostly due to the first common component. Poncela, Senra and Sierra (2014) extracted a principal component from 44 non-fuel commodity prices from 1992 to 2012. Evidence has also been found that commodity prices consistently display a tendency to revert towards one factor, see West and Wong (2014). The main drawback with extracting a single common component from commodities is that the estimated factor can be difficult to interpret and may not fully account for heterogeneity. Recent research has sought to decompose the sources of commodity price co-movement using more granular methods. For example, Yin and Han (2015) used a multilevel factor model to decompose 24 commodity returns into global, sectoral and idiosyncratic components. They highlighted the heterogeneous impacts of sectoral factors at different points in time.

A parallel and lively debate also spurred by the recent price boom-bust cycle, has focused on the determinants of commodity prices. First, it is

(2008, 2014), Neftci and Lu (2008), Xu and Ouenniche (2012), Ghoshray, Kejriwal and Wohar (2014) and Ouenniche, Xu and Tone (2014).

3 The Prebisch–Singer hypothesis examines whether the terms of trade of commodity exporters are trending, such that living standards would be further impoverished by specialising in commodity extraction with a secular decline in commodity prices. The hypothesis was revisited recently by Harvey *et al.* (2010).

frequently argued that commodity prices are primarily driven by global economic activity (e.g. Svensson, 2008; Wolf, 2008; Kilian, 2009; Abhyankar, Xu and Wang, 2013). This argument is related to increases in demand due to the unexpectedly strong economic growth in emerging economies after 2000 and their rapid recovery from the global financial crisis (Singleton, 2013; Frankel, 2014). The impact on commodity prices of rapid Chinese economic growth can similarly be understood as a demand shock. This has been highlighted in work by Dwyer, Gardner and Williams (2011), Roache (2012) and Frankel (2014) and is especially topical since recent commodity price collapses are related to fears about China's growth slowdown.

Monetary policy believed to have played an important role for commodity prices. For instance, Barsky and Kilian (2004) pointed out that high prices for oil and other commodities in the 1970s was because of an expansionary monetary policy. Frankel (2008, 2014) argued that a substantial increase in US real interest rates drove commodity prices down in the early 1980s. According to asset pricing models, commodity prices are determined by the expected discount rate and expected future returns.[4] Therefore, an increase in real interest rates will raise the discount factor, so the present value of future returns will fall and subsequently lead to lower commodity prices. Higher rates of return on fixed income assets will also offer substitution opportunities and reduce speculative demand for commodities. In addition, high interest rates increase the supply for storable commodities by increasing extraction incentives today, and/or by decreasing firms' desire to carry inventories. Gruber and Vigfusson (2012) showed that lower interest rates can reduce commodity prices' volatilities and interest rates can impact commodities heterogeneously.

Uncertainty may also be important for commodity prices. According to the standard option theory, given investment in primary commodities is irreversible (or at least partially irreversible), an increase in the variance of the distribution of investment returns would increase the option value of waiting for new information, causing delays in such investment and possible price effects (Dixit and Pindyck, 1994). Consequently, uncertainty will play a negative role in effecting the price of commodities. Beck (2001), for example, found some evidence on the relationship between risk and agricultural commodities. Byrne, Fazio and Fiess (2013) suggested a role for macroeconomic uncertainty in affecting the movements of commodity prices. While uncertainty may negatively affect commodity prices by encouraging investors to delay investments, it may also lead producers to lower their supply and production level and push prices up (Alam and Gilbert, 2017). Hence, it becomes an empirical question as to which uncertainty effect dominates.

In reviewing the literature, we have identified several empirical studies on investigating the determinants of commodity price fluctuations. For example, Vansteenkiste (2009) showed that the commodity common factor was affected by oil price, global demand, the US dollar effective exchange rate

4 If we assume commodity prices are determined like other assets, see Svensson (2008).

and the real interest rate. Using a FAVAR approach, Byrne, Fazio and Fiess (2013) related the common factor in commodity prices to their macroeconomic fundamentals. They found real interest rate and uncertainty were both negatively related to the common factor. Poncela, Senra and Sierra (2014) also applied a FAVAR to assess the impact of real interest rates, the US real effective exchange rate, VIX, world industrial production and an energy index. They found uncertainty after 2003 has played a more important role in explaining price fluctuations than real fundamentals, such as the real exchange and the real interest rate. West and Wong (2014) computed the first principal component correlation with fundamentals and found that commodities were positively related to industrial production and negatively related to the exchange rate. Alam and Gilbert (2017) extracted a common factor using principal components and found that monetary policy, demand and the US exchange rates significantly affected agricultural commodity prices. A common feature of all these empirical studies is that they rely on time-invariant regressions. Therefore, the impact of demand shocks on the commodity prices is assumed to be time-invariant or unconditional.

To summarise, our study extends the literature on the co-movement and determinants of primary commodity prices for the following reasons. While previous studies focused on a single common factor in a wide range of commodities, we account for heterogeneity by employing a dynamic hierarchical factor model to decompose commodities into global, sectoral and idiosyncratic components. Next, using a time-varying parameter factor augmented vector autoregressive model with stochastic volatility, we examine determinants of commonalities over time and across sectors. We now turn to formally laying out our econometric methodology.

3. Methodology

3.1. Dynamic hierarchical factor model

We adopt a four-level dynamic hierarchical factor model (DHFM) following Moench, Ng and Potter (2013). Let $X_{b,s,n,t}$ be commodity price series $(n = 1, \cdots, N)$, at each period $t = 1, \cdots, T$, in a given sub-sector s $(s = 1, \cdots, j)$ of sector b $(b = 1, \cdots, i)$ has four sources of variations: commodity-specific, sub-sector $(H_{b,s,t})$, sector $(G_{b,t})$ and common (F_t). Our dynamic factor model is set out as follows:

$$X_{b,s,n,t} = \lambda_H(L) H_{b,s,t} + e_{X,t} \tag{1}$$

$$H_{b,s,t} = \lambda_G(L) G_{b,t} + e_{H,t} \tag{2}$$

$$G_{b,t} = \lambda_F(L) F_t + e_{G,t} \tag{3}$$

where λ_H, λ_G and λ_F denote parameters for sub-sector factors, sector-factors and common factors, respectively. Error terms $e_{X,t}$, $e_{H,t}$ and $e_{G,t}$ denote commodity-specific, sub-sector and sector-level residual variations, respectively.

Note that $e_{X,t}$, $e_{H,t}$, $e_{G,t}$ and F_t are assumed to be stationary, normally distributed autoregressive processes of order one, AR(1), and evolve as follows:

$$e_{X,t} = \psi_X e_{X,t-1} + \varepsilon_{X,t} \tag{4}$$

$$e_{H,t} = \psi_H e_{H,t-1} + \varepsilon_{H,t} \tag{5}$$

$$e_{G,t} = \psi_G e_{G,t-1} + \varepsilon_{G,t} \tag{6}$$

$$F_t = \psi_F F_{t-1} + \varepsilon_{F,t} \tag{7}$$

where ψ_X, ψ_H, ψ_G and ψ_F denote the coefficient of the AR(1) dynamics. $\varepsilon_{X,t}$, $\varepsilon_{H,t}$, $\varepsilon_{G,t}$ and $\varepsilon_{F,t}$ follow $N(0,\ \sigma_j^2)$, $j = X$, H, G and F, which are uncorrelated across time and sectors. We assume that the factor loading matrix is constant and estimate one common factor per sector, and one factor per sub-sector. A standard method to estimate latent factors from a large number of data series is principal component analysis. However, principal component analysis would not account for potential relations between common and sector factors, nor the AR(1) time series structure described in equations (4)–(7). Moench, Ng and Potter (2013) propose a Markov Chain Monte Carlo (MCMC) method to overcome these problems. Following Moench, Ng and Potter (2013), we first employ principal components to obtain the initial values of factors, then run the MCMC method and discard the first 20,000 draws as burn-in and save every 100th of the remaining 50,000 draws.[5]

3.2. Factor augmented vector auto regression (FAVAR) models

Now we set out the basic Bayesian Factor Augmented VAR model to examine the determinants of commodity commonalities and explain how it can be extended to a time-varying parameter (TVP) model. The TVP-FAVAR model with stochastic volatility allows us to understand how changes in macroeconomic fundamentals affect real commodity prices over time.

3.2.1. Bayesian FAVAR model

The basic Bayesian FAVAR model can be written as follows:

$$AY_t = \sum_{i=1}^{p} \Gamma_i Y_{t-i} + u_t,\ \ t = p + 1, \ldots, T \tag{8}$$

where Y_t is a $K \times 1$ vector of endogenous variables and divided into two blocks: the first block includes the growth rate of real US industrial production ($Demand_t$), the real interest rate (R_t) and a risk term ($Risk_t$); the second

5 We followed the recommended number of iterations provided by Moench, Ng and Potter (2013). The detailed estimation procedures for the dynamic hierarchical factor model are reported in Appendix C.1 (in supplementary data at *ERAE* online)..

block includes the common factors in commodity prices (F_t);[6] Γ_i is a $K \times K$ matrix of coefficients, A is a $K \times K$ matrix of contemporaneous coefficient of Y_t, and u_t captures the structural shocks in the commodity market and macroeconomic conditions. We assume u_t to be i.i.d. and $N(0, \ \Sigma\Sigma)$. The lag length is two (i.e. $p = 2$),[7] where Σ is the diagonal matrix:

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_k \end{pmatrix}$$

To specify the simultaneous relations of the structural shock, we employ its reduced-form representation by multiplying both sides by A^{-1}, resulting in:

$$Y_t = \sum_{i=1}^{p} B_i Y_{t-i} + A^{-1}\Sigma\varepsilon_t, \ \ \varepsilon_t \sim N(0, I_k) \tag{9}$$

where $B_i = A^{-1}\Gamma_i$ for $i = 1,\ldots,p$. We can stack all the VAR coefficients (B_i) into a $K^2p \times 1$ vector to form B and define $X_t = I_k \otimes (Y'_{t-1}, \ldots, Y'_{t-p})$, where $\otimes$ denotes the Kronecker product. We rewrite equation (9) as:

$$Y_t = X_t B + A^{-1}\Sigma\varepsilon_t \tag{10}$$

Note that the reduced-form residuals ε_t are correlated between each equation and can be viewed as a weighted average of the structural shocks u_t in equation (8). In order to orthogonalise the shocks, we impose a recursive structure on the contemporaneous terms and assuming that A is lower-triangular,

$$A = \begin{pmatrix} 1 & \cdots & \cdots & 0 \\ a_{21} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ a_{k1} & \cdots & a_{k,k-1} & 1 \end{pmatrix}$$

The ordering of the variables is as follows: $Y_t = [\,Demand_t, \ R_t, \ Risk_t, \ F_t]$. The structural shocks u_t are identified by decomposing the reduced-form errors ε_t as follows:

6 When accounting for commodity heterogeneity we replace the common factor F_t with sectoral factors $G_{b,t}$ ($Sectoral_{b,t}$).

7 Most lag length specification tests (e.g. Final Prediction Error; Akaike Information Criterion; and Hannan-Quinn Information Criterion) suggest that two lags should be included for our model with quarterly data.

$$\varepsilon_t = \begin{pmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{R} \\ \varepsilon_t^{Risk} \\ \varepsilon_t^{F} \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a_{21} & 1 & 0 & 0 \\ a_{31} & a_{32} & 1 & 0 \\ a_{41} & a_{42} & a_{43} & 1 \end{bmatrix} \begin{pmatrix} u_t^{Demand} \\ u_t^{R} \\ u_t^{Risk} \\ u_t^{F} \end{pmatrix} \tag{11}$$

In the global macroeconomic block, u_t^{Demand} denotes the *aggregate demand shock* that captures the shift in the demand for all commodities driven by the global business cycle. Next, u_t^R is the *real interest rate shock* that reflects deviations from the expected or average monetary policy, whether via changes in the nominal interest rate, expected inflation, or both (e.g. Frankel, 2008). Finally, u_t^{Risk} denotes the *risk shock* that captures innovations to agents' expectations about future economic and financial conditions. Intuitively, this can also be thought of as precautionary trading arising from revisions to commodity market expectations. These expectations may arise from elevated risks in financial markets, as in the global financial crisis. In the commodity market block, u_t^F is the shock to the common factor in commodity prices.

The restrictions on A^{-1} are based on the following assumptions and economic intuition. The first assumption is that global real economic activity does not respond immediately to real interest rate shocks, uncertainty shocks and commodity markets, but does so with a delay of at least a quarter. Our second exclusion restriction imposed in the VAR is that increases in uncertainty regarding the economic and financial outlook will not affect real interest rates immediately. Our third assumption is that innovations to risk respond to demand and policy shocks without a delay. These assumptions are based on Bernanke, Boivin and Eliasz (2005) and Stock and Watson (2005), who have classified the series into slow-moving variables, such as real output and income, wages and spending; and fast-moving variables such as stock prices, money and credit. Global real economic activity is considered slow-moving: it is plausible that consumers and firms slowly revise their spending plans after a monetary policy or financial market shocks. In contrast, commodity price volatility is considered as a fast-moving variable, responding contemporaneously to slow-moving variables and policy shocks.[8] For the commodity market block, we assume that u_t^F does not affect global real activity, the real interest rates and uncertainty within a given quarter, but instead with a delay of at least one quarter. This is imposed through the exclusion restrictions in the last column of A^{-1}. This assumption is implied by the standard approach of treating innovations to the price of commodities as predetermined with respect to the economy (e.g. Kilian and Park, 2009).

8 A similar identification strategy in commodity price literature is also followed by Akram (2009), Lombardi et al. (2012), Byrne et al. (2013), Hammoudeh et al. (2015), and Alam and Gilbert (2017)'s first identification strategy.

We estimate the FAVAR model in the context of Bayesian inference and adopt the independent Normal-Wishart prior, which is more flexible than the natural conjugate prior. The prior distributions are described as:

$$B \sim N(\underline{B},\ \underline{V_B})$$

$$\Sigma^{-1} \sim W(\underline{S}^{-1},\ \underline{\nu})$$

Where $\underline{B} = 0$, $\underline{V_B} = 10I_4$, $\underline{S} = I_4$, and $\underline{\nu} = 5$ as in Koop and Korobilis (2010). The conditional posterior distributions $p(B|Y, \Sigma^{-1})$ and $p(\Sigma^{-1}|Y, B)$ are computed by the MCMC method. Following Primiceri (2005), we use a training sample prior to obtain the initial Σ^{-1}. The training sample is the first 40 observations (1974: Q1 to 1983: Q4). Using the MCMC method, 20,000 samples are obtained after the initial 10,000 samples are used as burn-in and discarded.[9]

3.2.2. Time-varying parameter FAVAR with stochastic volatility

Note that all parameters in equation (10) are time-invariant. Next, we adjust the model by allowing these parameters to vary over time, as follows:

$$Y_t = X_t B_t + A_t^{-1} \Sigma_t \varepsilon_t \tag{12}$$

where the coefficients B_t, and the parameters A_t, and Σ_t are all time varying. Time-varying parameters allow the relationship between fundamentals and commodities to evolve over time. Stochastic volatility allows for varying shock intensity and improves estimation precision (see Nakajima, Kasuya and Watanabe, 2011). We follow Primiceri (2005) and let $a_t = (a_{21}, a_{31}, a_{32}, a_{41}, a_{42}, a_{43})'$ be a stacked vector of the lower-triangular elements in A_t and $h_t = (h_{1,t}, \ldots, h_{k,t})'$ with $h_{j,t} = \log\sigma_{j,t}^2$, for $j = 1, \ldots, k$ and σ_{jt} is the diagonal element of Σ_t. We assume that the parameters in (12) follow a driftless random walk process, thus allowing both temporary and permanent shift in the parameters:

$$\begin{array}{l} B_t = B_{t-1} + u_{B,t}\ , \\ a_t = a_{t-1} + u_{a,t}\ , \\ h_t = h_{t-1} + u_{h,t}\ , \end{array} \begin{pmatrix} \varepsilon_t \\ u_{B,t} \\ u_{a,t} \\ u_{h,t} \end{pmatrix} \sim N\left(0, \begin{pmatrix} I_K & 0 & 0 & 0 \\ 0 & \Sigma_B & 0 & 0 \\ 0 & 0 & \Sigma_a & 0 \\ 0 & 0 & 0 & \Sigma_h \end{pmatrix}\right),\ t = 1, \ldots, T$$

The shocks to the innovations of the time-varying parameters are assumed uncorrelated among the parameters B_t , a_t and h_t. We further assume for simplicity that Σ_B, Σ_a and Σ_h are all diagonal matrices. Our dynamic specification

9 Note that for Bayesian VAR and TVP-FAVAR models, we found that 10,000 burn-in and 20,000 sample draws are sufficient for the MCMC algorithm to converge. As a robustness check, we have employed the consistent draw number were used in the dynamic hierarchical factor model (20,000 burn-in and 50,000 sample draws). We find that the change of draws does not affect the estimation results – see Figure B12 (Appendix in supplementary data at *ERAE* online).

permits the parameters to vary and the shock log variance follows a random walk process to capture possible gradual or sudden structural changes, as discussed by Primiceri (2005).

For estimation, we employ a training sample prior, as shown in Section 3.2.1 and the prior distributions are set as follows:

$$B_0 \sim N(B_{OLS}, 4 \cdot \mathrm{V}(B_{OLS}))$$

$$A_0 \sim N(A_{OLS}, 4 \cdot \mathrm{V}(A_{OLS}))$$

$$h_0 \sim N(h_{OLS}, 4 \cdot I_k)$$

where B_{OLS}, A_{OLS} and h_{OLS} denote the OLS point estimates and $\mathrm{V}(\cdot)$ denotes the variance. We also need to set the hyper-parameters Σ_B, Σ_a and Σ_h and we postulate the following inverse-Wishart prior distributions:

$$\Sigma_B \sim IW(k_B^2 \cdot 40 \cdot \mathrm{V}(B_{OLS}), 40)$$

$$\Sigma_a \sim IW(k_a^2, 2)$$

$$\Sigma_{1,h} \sim IW(k_h^2 \cdot 2 \cdot V(A_{1,OLS}), 2)$$

$$\Sigma_{2,h} \sim IW(k_h^2 \cdot 3 \cdot V(A_{2,OLS}), 3)$$

$$\Sigma_{3,h} \sim IW(k_h^2 \cdot 4 \cdot V(A_{3,OLS}), 4)$$

where $k_B = 0.01$, $k_\alpha = 0.1$ and $k_h = 1$; $\Sigma_{1,h}$, $\Sigma_{2,h}$ and $\Sigma_{3,h}$ denote the three blocks of Σ_h and $A_{j,OLS}$ for $j = 1, \ldots, 3$, denotes the three corresponding blocks of A_{OLS}. The estimation procedure is the MCMC method and the first 10,000 samples are discarded, and 20,000 samples are obtained for the inference.[10]

4. Data

In this paper, we use quarterly primary commodity prices and fundamentals data from 1974Q1 to 2014Q3. For the dynamic hierarchical factor model, we collect a panel of 38 commodity prices from the International Monetary Fund (IMF) *International Financial Statistics* and World Bank commodity price data. Following the structure of the IMF non-fuel commodity index, we arrange the commodity data into three sectors: (i) *agricultural raw materials*, (ii) *food* and (iii) *metals*. First, the materials sector includes eight commodities: cotton, hides, plywood, rubber, hardwood logs, hardwood sawn wood, coarse wool and fine wool. Second, our dataset includes 23 food commodities, and we further decomposed them into four sub-sectors; namely, cereals

10 The details of the MCMC procedure for TVP-FAVAR model are reported in Appendix C.2 (in supplementary data at *ERAE* online). (Primiceri, 2005; Koop and Korobilis, 2010; Nakajima, Kasuya and Watanabe, 2011).

Table 1. Data and model structure

Sector	Sub-sector	N
Food	Cereals	5
	Meat	3
	Vegetable oil and protein meals (non-fuel oil)	9
	Others	6
Materials		8
Metals		7
Total		38

Notes: This table summarises the different sectors in our data set. These are food, agricultural raw materials and metal sectors. Sub-sectors for real commodity prices are also included. *N* denotes the number of series in each sector/sub-sector.

(i.e. barley, maize, rice, sorghum and wheat); meat (i.e. beef, lamb and chicken), vegetable oil and protein meals (i.e. coconut oil, copra, groundnuts, groundnut oil, linseed oil, palm oil, soybeans, soybeans meal, and soybeans oil) and others (i.e. cocoa beans, coffee, tea, banana, fishmeal, sugar). Finally, we include seven metal commodities: aluminium, copper, gold, lead, silver, tin and zinc. Table 1 summarises our data and model structure of commodity prices. Quarterly data is preferred when estimating our time-varying parameter model: TVP-FAVAR estimation at a monthly frequency would require many lags to capture data dynamics, and hence would be computationally intensive (Nakajima, Kasuya and Watanabe, 2011). Before the factor analysis, we deflate the nominal commodity prices using US CPI. Before the factor analysis, we deflate the nominal commodity prices using US CPI. Note that our factor model requires all the data are stationary, hence we employed the Augmented Dickey–Fuller (ADF), Phillips–Perron (PP) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests to ascertain whether real commodity prices are stationary (Stock and Watson, 2009; Moench, Ng and Potter, 2013). The null hypotheses for ADF and PP are the existence of a unit root I(1), so if the series is stationary I(0), the ADF and PP tests should reject the null hypothesis. In contrast, the null hypothesis of the KPSS statistic is that the series is stationary, i.e. I(0). In terms of the processes under study, we have found evidence that the level of real commodity prices contains a unit root. Hence, we have taken the logarithmic transformation for all real commodity prices series. A detailed description of the commodity price series is presented in Table A1 (Appendix in supplementary data at *ERAE* online).

Next, we gather macroeconomic fundamentals data from the Federal Reserve Bank of St. Louis and Datastream. We first use US industrial production as our proxy for global economic activity ($Demand_t$). The rationale for using this proxy is that the growth of industrial production will reflect changes in the demand for industrial commodities (e.g. copper, lumber) and it will also impact demand for non-industrial commodities (e.g. cocoa, wheat) as income changes, see Pindyck and Rotemberg (1990). Given the potentially increasing importance of the Chinese economy for commodities, we also proxy demand using the growth rate of Chinese industrial production.[11] Second, we consider the role of the real short-term interest rate based on the three-month US Treasury Bill as a proxy for monetary policy shocks (e.g. Primiceri, 2005). The real rate (R_t), is obtained by subtracting US CPI inflation from the nominal interest rate, based on the Fisher equation. Note that we also test the robustness of our approach by using the federal fund rate as an alternative interest rate proxy (e.g. Bernanke, Boivin and Eliasz, 2005; Frankel, 2014).

In order to examine the risk effects on commodity prices' growth, we model commodity markets' risks that arise from agents' perspectives on future outcomes ($Risk_t$). To that end, we fit a GARCH model of the log difference of the daily S&P GSCI (Goldman Sachs Commodity Index) to cover the period from January 1970 to September 2014.[12] We should note that prior to estimating this model, we confirmed the presence of ARCH effects in the GSCI using the Lagrange Multiplier test. We also check whether the standardised residuals exhibit higher order autocorrelation and ARCH effects. Ascertaining that the selected model is well specified, we take the within quarter average of the estimated conditional variances to match the frequency of the commodity data. This series is then used as a measure of uncertainty in the market. Here, higher levels of conditional variance imply higher perceived uncertainty. In such an environment, decision makers (e.g. fund managers and commodity producers) will not be able to predict the viability nor returns of projects. Thus, one may behave more conservatively and opt to postpone the investment to avoid potentially large losses when the project outcome is unfavourable. To check for the robustness of our investigation, we also proxy uncertainty using the within quarter standard deviation of the GSCI non-energy spot index and a GARCH model fitted to the stock market. Using ADF, PP and KPSS tests, we test for all series involved including macroeconomic variables, common and sectoral factors. We find that the real US industrial production rate and the real Chinese industrial production rate are non-stationary in levels, and we have used the first difference of these

11 As a further robustness check, we consider Kilian's (2009) global economic activity index as an alternative proxy for demand. Note that Kilian's series is a measure from an equal-weighted index of the percentage growth rates of a panel of single dry cargo ocean shipping freight rates in dollars per metric ton. Given the supply of ocean-going vessels is likely to be inelastic in the short-run, shipping rates shall reflect global demand for commodities. One of advantages of the Kilian index is that it can be constructed as far back as 1968, the primary data source underlying the Kilian index is the report on 'Shipping Statistics and Economics' published by *Drewry Shipping Consultants Ltd*. The Kilian index can be found from Lutz Kilian's homepage: http://www-personal.umich.edu/~lkilian/reaupdate.txt. Kilian and Zhou (2018) address several potential objections that has been raised in the literature e.g. failing to consider the cost of bunker fuel; the shipbuilding cycle (e.g. Odom, 2010; Ravazzolo and Vespignani, 2017). They also provide complementary evidence that supports the existence of a persistent global economic slowdown between 2010 and 2016. We thank an anonymous referee for this point.

12 For standard references to GARCH estimation see Engle (1982) and Bollerslev (1986).

variables to remove the unit root.[13] We next proceed to estimate our factor and time-varying empirical models.

5. Empirical results

This section presents our core results on the nature and determinants of commodity prices. We first report the empirical findings on commodity price co-movement using the dynamic hierarchical factor model of Moench, Ng and Potter (2013). Next, we discuss the results from a constant parameter FAVAR model and those from a time-varying parameter FAVAR model with stochastic volatility.

5.1. Commodity price co-movement

The four-level dynamic hierarchical factor model allows us to capture the aggregate common dynamics, as well as to track the developments in different commodity sectors. Figure 1 plots the estimated global common factor, $\hat{F}$, plus material, food and metal sectoral factors, $\hat{G}$. Similarities

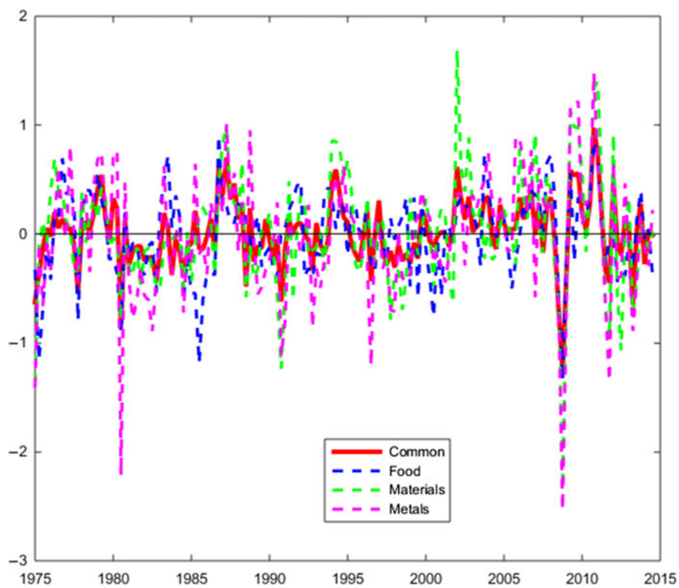

Fig. 1. Commonalities in commodity prices. *Notes*: The graph presents estimated aggregate common factor (Common), along with sectoral factor for the returns in food (Food), agricultural raw materials (Materials) and metal (Metals). The sectoral commonalities are identified dynamic hierarchical factor model in equations (1)–(3), see Moench, Ng and Potter (2013).

13 To save space, we do not report these test results. They are available on request.

between the sectors include, for instance, the commodity price peaks in mid-2008, fall thereafter and peak again in 2011. There are some important differences between the common and sectoral factors, especially during episodes of extreme price volatility. For example, agricultural raw material and metal sectors react more strongly than the common factor during the striking collapse of commodity prices in the global financial crisis. The growth of materials and metals also exceeds the common factor before the crisis. This may imply that agricultural raw materials and metals are more sensitive to the global business cycle, compared to food commodities.

We now turn to the important question of evaluating how much of the variation in real commodity prices can be attributed to the aggregate common ($Share_F$), sector ($Share_G$) and sub-sector ($Share_H$) components and to idiosyncratic noise ($Share_Z$). Table 2 reports the average posterior means (μ) and standard deviations (σ) of the estimated variance shares for the four-level decomposition of our data set, including all sectors and sub-sectors. Our first evidence of heterogeneity in commodities is that the aggregate common factor is most important for metals in which a third of commodity price variation is explained by the global commonalities in commodities, while less than 1 per cent is explained for food-meats. Mirroring the latter result, idiosyncratic variations play an important role for food-meats as it accounts for most of the variation in this sub-sector (i.e. $\mu = 0.742$). On the other hand, cereals, non-fuel oils, agriculture raw materials and metals are explained to a lesser extent by idiosyncratic variation. As mentioned above, our hierarchical model goes beyond principal components since it accommodates sector and sub-sector

Table 2. Variance decomposition of commodity prices

Sector	Sub-sector	Global [$Share_F$]	Sector [$Share_G$]	Sub-sector [$Share_H$]	Idiosyncratic [$Share_Z$]
Food	Cereals	0.215	0.177	0.183	0.424
		(0.017)	(0.013)	(0.013)	(0.033)
	Meat	0.007	0.006	0.245	0.742
		(0.012)	(0.010)	(0.170)	(0.182)
	Non-fuel oil	0.243	0.200	0.037	0.520
		(0.028)	(0.024)	(0.006)	(0.045)
	Others	0.024	0.020	0.090	0.866
		(0.016)	(0.013)	(0.044)	(0.063)
Materials		0.207	0.096		0.697
		(0.052)	(0.029)		(0.059)
Metals		0.330	0.098		0.572
		(0.062)	(0.023)		(0.069)

Notes: This table summarises our four-level variance decomposition of real commodity price based on a dynamic hierarchical factor model (DHFM) presented in equations (1)–(3). The four levels are the global factor [$Share_F$], sector [$Share_G$], sub-sector [$Share_H$] and idiosyncratic shocks [$Share_Z$]. We report the means and in parentheses standard deviations of the estimated variance shares for the model's four levels, including all sectors and sub-sectors of our data set. The sample period is 1974Q1 to 2014Q3.

level shocks. In sum, our result suggests that co-movement in commodity prices co-exist with heterogeneous variations between sectors and highlight the importance of modelling common variations at different levels (Moench, Ng and Potter, 2013; Yin and Han, 2015). Given these results, sectoral heterogeneity is the focus of our later analysis.

5.2. Commonalities, fundamentals and heterogeneity

In this section, we model commodity prices by examining the relationship between their common factors and key macro fundamentals highlighted in the literature. We first present impulse response functions to the shocks, based on a Bayesian FAVAR model with common ($\hat{F}$). Thereafter, we investigate whether FAVAR models are robust to time variation and commodity heterogeneity, utilising findings from our TVP-FAVAR model for the common factor ($\hat{F}$) and groups of commodities ($\hat{G}$). As we noted above, such an approach has not been extensively researched in the literature.

5.2.1. Impulse responses from a FAVAR model

Using our Bayesian FAVAR model, Figure 2 depicts impulse responses of the common factor of commodity returns to one-standard deviation three macro shocks (i.e. demand, real interest rate and risks) over three sample periods. That is the full sample period from 1974Q1 to 2014Q3; and two subsamples 1974Q1 λ 1993Q4 (period 1) and 1994Q1 λ 2014Q3 (period 2). We present results for a ten-quarter response horizon. Our responses include the posterior median as the solid line, while the dotted lines are the 16th and 84th percentiles of the posterior distribution.[14]

We start by reporting the full sample period results in first column of Figure 2. First, we find that demand shocks, as measured by US industrial production growth, lead to an immediate increase in the real price of commodities, but the effect declines sharply after four quarters – see the top left figure in Figure 2. We also see that commodities' responses to demand are important for the first four quarters, since the zero axis is not within the error bands. Our finding is consistent with Boughton and Branson (1991); Vansteenkiste (2009); Byrne, Fazio and Fiess (2013) and West and Wong (2014), who also find that positive innovations to measures of the global business cycle positively impact the price of commodities. For the full sample period, we find that an unexpected increase of 150 basis points (one standard deviation) in the interest rate (a contractionary policy) leads to an immediate and sizable decrease in commodity prices for the first quarter as shown in the middle-left window of Figure 2. This is in line with Scrimgeour (2014) and Alam and Gilbert (2017). Furthermore, an uncertainty shock causes an immediate drop in commodity prices and then dies out five quarters after the shock. See the bottom-left response in Figure 2. Therefore, our

14 Under normality, the 16th and 86th percentiles correspond to the bounds of one standard-deviation (Primiceri, 2005).

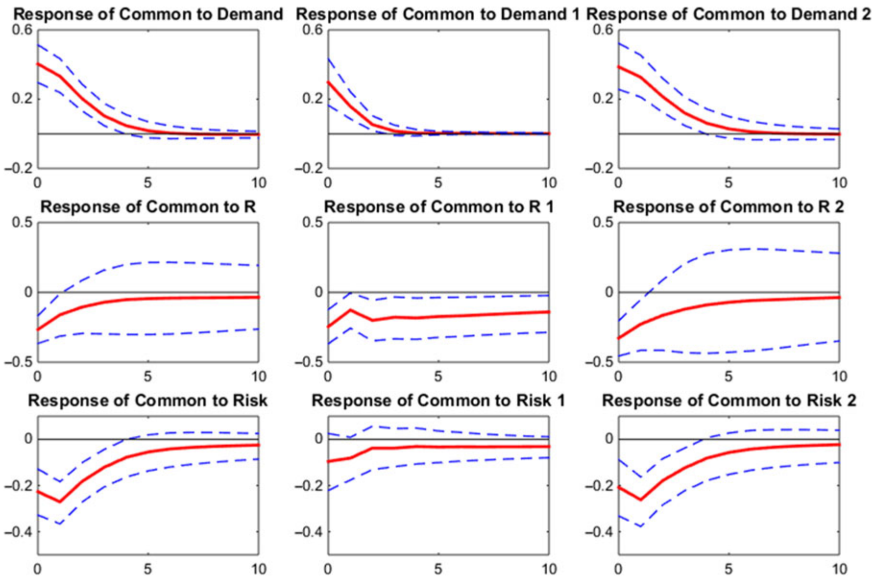

Fig. 2. FAVAR impulse responses of common factor. *Notes*: The nine graphs in the figure plot the median responses of common factor (solid line) to each of the three macro shocks that affect the commonalities in commodity returns for three sample periods. Note that the first column reports the full sample period (1974Q1 λ 2014Q3) result, and the second and third columns report two sub samples; 1974Q1 λ 1993Q4 (i.e. period 1) and 1994Q1 λ 2014Q3 (i.e. period 2). We also provide the 16th and 84th percentile error bands in dashes. The common factor is extracted from the DHFM equation (3).

findings not only confirm the view of Frankel (2008), and Byrne, Fazio and Fiess (2013) that interest rates have an adverse impact on commodity prices, but also are consistent with the idea of Beck (1993), Dixit and Pindyck (1994) and Kulatilaka and Perotti (1998) that risk is strongly associated with movements in commodity prices.

Next, we consider whether the impact of fundamental shocks is broadly time varying, by splitting the sample into two sub-periods. The second and third column of Figure 2 displays the median impulse responses of the common factor to macro shocks over the subsamples 1974Q1–1993Q4 and 1994Q1–2014Q3, respectively. We identify an evolving relationship between primary commodity prices and macro fundamentals. To be more specific, the response of commodity prices to the one standard deviation contractionary monetary policy shock is more persistent in the first subsample, while the reaction to demand and uncertainty shocks is more pronounced in the second subsample.

In addition, we have replaced the common factor with three sectoral factors and apply our three macroeconomic shocks. We have observed the important difference in sectoral responses to macroeconomics fundamentals over the full sample period – see Figure B1 (Appendix in supplementary data at *ERAE* online) for details.[15]

5.2.2. TVP-FAVAR model with stochastic volatility

The previous results were based on the assumption that the impact of fundamentals on commodity prices was time-invariant. Our subsample analysis, with exogenously identified sub-periods, indicates that this assumption is open to question, hence we now adopt more flexible methods.

In this section, we focus on the time evolution of the relationship between commonalities in commodity returns and macroeconomics shocks using a TVP-FAVAR model with stochastic volatility. Such an approach allows us to consider the evolving impact of Chinese demand, recent monetary policy and the role of risk during the financial crisis. Note that for a standard VAR model whose parameters are time-invariant, we can graph one impulse response profile for each shock – see Figure 2. For the time-varying parameter models, however, there will be a different set of coefficients in every time period. So, we will have a different impulse response at each point in time. Although one can draw a three-dimensional plot for the time-varying impulse responses, it is common practice to present the impulse responses for a selected horizon over time and/or at a selected point in time (e.g. Primiceri, 2005; Koop and Korobilis, 2010; Nakajima, Kasuya and Watanabe, 2011; Baumeister and Peersman, 2013). Therefore, we plot both the impulse responses for the full sample period, and for up to a ten-quarter horizon for three specific time periods.

Figure 3 graphs contemporaneous time-varying impulse responses of the global commodity factor to one standard deviation increases in demand, real interest rates and risk. In this figure, the posterior median is the solid line and the dotted lines are the 16th and 84th percentiles of the posterior distribution. The top panel of Figure 3 shows that aggregate demand has a consistently important and positive relationship with commodity prices commonalities since the zero axis is below the 68 per cent posterior credible interval. This is reasonable because an expansion in economic activity shall increase industrial commodity demand and drive up commodities prices (see, Kilian, 2009; Frankel, 2014; Byrne, Lorusso and Xu, 2018). The effect of aggregate demand shocks on real commodity prices is evidently time varying, as we observe a larger response in the 2000s compared to the early years of our sample. However, this positive effect becomes smaller after the global financial crisis. This corresponds with the unexpected increase in demand in the 2000s from many emerging market economies, such as China and India, as they became more prominent in the world trade of commodities, while the global financial crisis triggered recessions across many countries that led to a significant decline in demand (e.g. Wolf, 2008; Kilian, 2009).

15 We have also carried out the same exercise for all three sectoral factors over two sub sample periods and we observed evolving relationships between sectoral commonalties and macro fundamentals. These results are available on request.

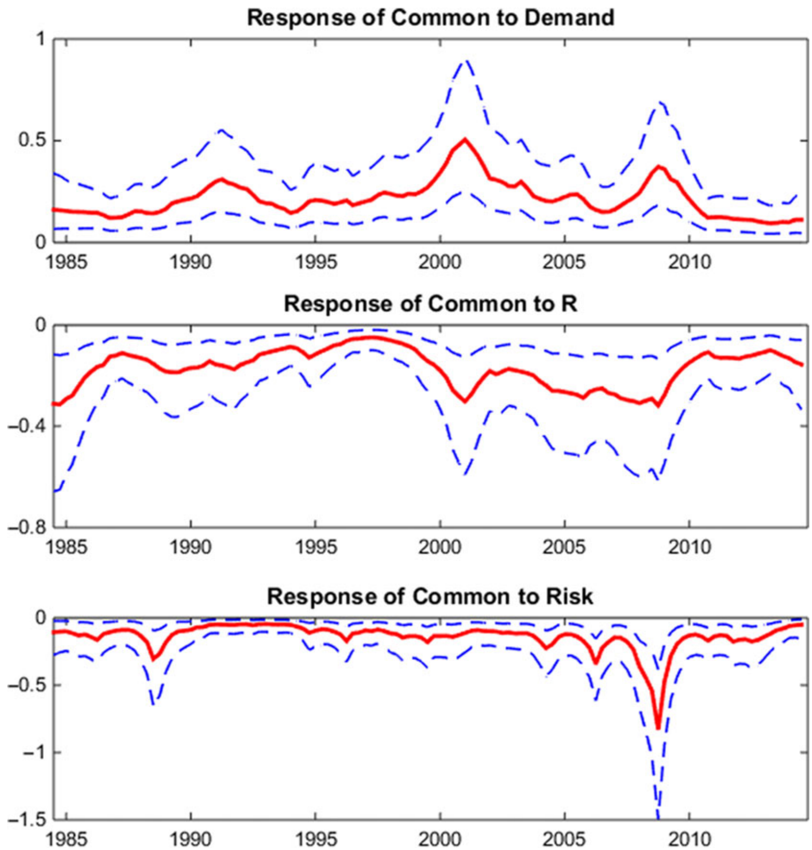

Fig. 3. TVP-FAVAR common factor response. *Notes*: The three graphs in figure plot the median impulse responses of the common factor (solid line) to each of the three macro shocks that affect commodity returns. We also provide 16th and 84th percentile error bands with dashes. The estimates are based on the TVP-FAVAR model in equation (11). Each panel measures how a unit impulse of shocks impacts the commodity common factor over time. Here we use the full sample period.

Second, we find the commodity common factor responded negatively to a one standard deviation contractionary monetary policy shock in real interest rate – see the middle panel of Figure 3.[16] Note that real commodity prices tend to fall in response to higher real interest rate until commodities are 'undervalued', and future prices are expected to rise sufficiently to offset the higher interest rate. Only then shall firms and investors hold inventories, despite the high carrying costs (Frankel, 2008). This negative effect is persistent over the entire sample period and there is evidence of time variation in the response of the commodity prices, as the posterior intervals are narrow. The peak impact of real

16 Note that a policy induced monetary contraction can temporarily raise the real interest rate via a rise in the nominal interest rate, a fall in expected inflation or both.

interest rate shocks on commodity prices was during the 2000s. This was a period of rapid commodity price inflation and the Federal Reserve carried out aggressive expansionary policies to combat the dot.com bubble recession and kept interest rates low, which sowed the seed of an asset price bubble that eventually burst in 2008 (Hammoudeh, Nguyen and Sousa, 2015). In response to the global financial crisis, in December 2008, the Federal Reserve switched to the Zero Interest Rate Policy (ZIRP), a seven-year period in which the target range for the Federal Funds rate was pegged between 0 and 0.25 per cent. This was a period of low productivity growth and the real price of commodities become less elastic towards to monetary policy shocks. With the commencement of the ZIRP, the Federal Reserve had limited nominal interest rate ammunition to affect the economy in general and asset prices in particular. Our result of a weaker post-crisis impact from real interest rates on commonalities is consistent with this.

Using our more flexible time-varying parameter methodology, the third shock to the common factor that we consider is that of uncertainty. This uncertainty proxy is from the commodity market, as discussed above. We find uncertainty had a substantial negative impact on commodity commonalities, see the bottom panel of Figure 3. The acutely time-varying impact is closely associated with the Global Financial Crisis, in periods where market volatility rose to levels that have rarely been seen since the Wall Street Crash.[17] Recall that commodities also fell sharply when uncertainty rose during the crisis period – see Figure 1. We would like to point out that the median value of commodity commonality reaches 0.8 in 2008Q3, which is a much larger response than that for other shocks. For example, the median value of the impact generated by the one standard deviation contractionary monetary policy shock during that period was about 0.3. Hence, the Federal Reserve needs to cut the interest rate by two to three standard deviations to fully mitigate the negative effect on commodity price. Our finding is in line with periods of pegged federal funds between 0 and 0.25 per cent. Although still important, we observe a smaller impact of the uncertainty shock on commodities before and after the crisis in 2008. The latter reduction is consistent with the partial success of a variety of government actions that were implemented to promote the liquidity, solvency of credit markets and financial market stability. As noted earlier, uncertainty effects are known to be short-lived, and our time-varying results highlight that they are especially time specific. As many economies were slow to emerge from the effects of the global financial crisis, what we do not need is an increase in commodity market risk that would adversely affect the price of a wide range of commodities. Our results provide another reason to pay attention to commodity market stability.

17 Recall that, the reaction of subprime mortgages and securitised products raised serious concerns about the solvency and liquidity of financial institutions. This led in 2008 to a full-blown banking crisis following the failures of Lehman Brothers, and government takeovers of Fannie Mae, Freddie Mac, and AIG (Ivashina and Scharfstein, 2010).

5.2.3. Sector-specific commonalities in a TVP-FAVAR model

To account for heterogeneity in commodity prices, we now replace the common factor in our TVP-FAVAR model with three sectoral factors for agriculture raw materials, food and metals. The TVP-FAVAR impulse responses for material, food and metal sectors in Figure 4 share some similarities. For example, we find that aggregate demand shocks are positively related to the real price of commodities across all three sectors throughout most of the sample period, while real interest rate shocks and uncertainty shocks are negatively related to commodity prices for certain periods.

There are important differences, however, among these three sectors' responses to macro shocks that underscore heterogeneity in the commodity market. The metal sector generally responds more positively to the global business cycle than the material and food sector in Figure 4. One possible

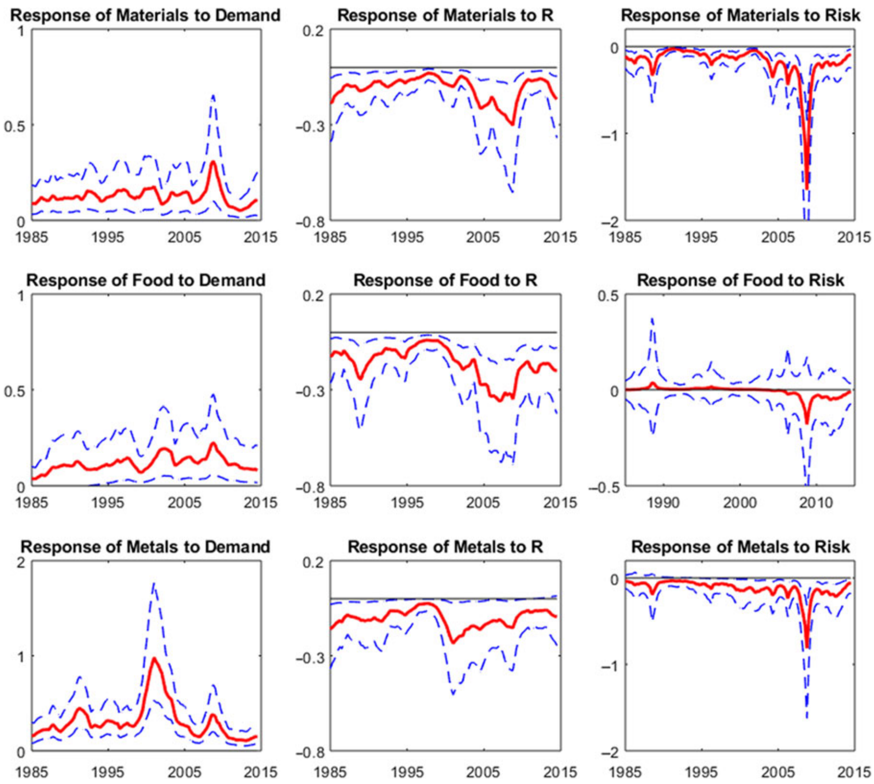

Fig. 4. TVP-FAVAR sectoral factor responses. *Notes*: The nine graphs in the figure plot the median responses of three sectoral factors ($Sectoral_{b,t}$) to each of the three macroeconomic shocks with 16th and 84th percentile error bands. The three shocks are to demand, the real interest rate and risk. The estimate responses are based on the TVP-FAVAR model in equation (11). Each panel measures how a unit impulse of shock impacts the commonalities of commodity prices over time. Here, we use the full sample period to consider the time-varying response.

explanation for this is that the unexpectedly rapid pace of industrialisation and urbanisation of emerging economies has dramatically increased the demand for metal commodities, such as copper, which are core materials in constructions and electronics. Issler, Rodrigues and Burjack (2014) also presents empirical evidence that cycles in metal prices are synchronised with the global economy. On the other hand, while greater economic activity without commensurate increases in population raises incomes, this is less likely to increase demand for food products overall and there may be substitution effects between different food commodities (Carter, Rausser and Smith, 2011).

However, we observe more homogeneous responses among material, food and metal sectors to demand after the crisis. This is consistent with the earlier literature on commodities moving together as an asset class (e.g. Byrne, Fazio and Fiess, 2013, and West and Wong, 2014). A number of authors study the cause of recent high food prices and increased cross-commodity linkages, and they provide evidence of a positive relationship between biofuel production and food prices, notably in the US (e.g. Mallory, Irwin and Hayes, 2012; Avalos, 2014). The growth in the subsidised biofuel industry raised concerns among stakeholders about global food shortages and food poverty.

In terms of real interest rate shocks, the zero axes are always out with the error bands for the agriculture raw materials and food, unlike for metals, suggesting that materials and food commodities are more sensitive to interest rate shocks than metals – see the second column of Figure 4. In particular, our sectoral evidence suggests that real interest shocks have a more persistent impact on food prices, possibly because monetary policy is more closely aligned with food prices than materials and metals. Furthermore, one of the key transmission mechanisms for changes in real interest rates to commodity price fluctuations is through the carrying cost of inventories. During high real interest rates periods, firms' desire to carry food inventories decreased faster compared to materials and metals. Food may be more interest-rate elastic compared to other commodities, since the physical costs of food inventory are already substantial and profit margins are therefore smaller. Our findings are consistent with Hammoudeh, Nguyen and Sousa (2015), who also found the effects of US monetary policy led to heterogenous responses from different types of commodity sector prices, which depend on the characteristics of particular commodity market (e.g. future price expectations, weather conditions, storability of commodity, easiness of supply and strength of demand for commodity inventories).

Turning to the time-varying impact of risk, our measure of commodity price uncertainty adversely affected the price of industrial commodities. This effect was particularly acute towards the end of 2007, see the third column of Figure 4 for the response of the three sectors to uncertainty. The global financial crisis caused a global recession and raised fears about future economic conditions. The heterogeneous responses are reasonable, since a large risk shock will reduce production activity and demand for raw materials and metals. Food, on the other hand, is more impervious to risk and the financial conditions more generally, possibly due to the importance of maintaining food consumption, even in a crisis.

To sum up our results so far, we provide empirical evidence that macro fundamentals affect the returns of a large panel of commodities and these effects vary over time. There is also important heterogeneity in the responses of the agricultural raw material, food and metal sectors.

5.3. China and the global commodity market

Several studies suggest that strong economic growth in China since 2000 has raised global demand for a broad range of commodities, and has resulted in increasing commodity prices (e.g. Kilian, 2009; Roache, 2012; Frankel, 2014). Again, prior studies assumed the impact of macro determinants on commodities has not changed over time. Therefore, we consider the extent to which commonalities in commodity prices are affected by unexpected increases in Chinese economic activity, accounting for heterogeneity and a potentially evolving relationship over time. We use the real growth rate of Chinese industrial production as a proxy for Chinese economic activity. We focus on the more topical post 2000 period, due partly to data availability.

Figure 5 plots time-varying impulse responses of the global commodity factor and three sectoral factors to the one standard deviation increase in Chinese demand immediately after the shock. The posterior median response is once again the solid line and dotted lines are the 16th and 84th percentiles of the posterior distribution. First, we find that unexpectedly strong demand from China led to a persistent increase in the common factor of commodities and this effect was more substantive in 2008 and 2012. The response of the common factor to Chinese demand is found in the top left panel of Figure 5. When we consider the response of commodity sectors to Chinese shocks, we again observe a strong positive impact on metal commodities due to Chinese demand. Materials are only more recently impacted by Chinese demand. Interestingly, we also find Chinese economic activity impacted food prices. In fact, greater real economic activity leads to higher level of incomes and rising incomes drive greater food consumption, particularly in developing countries in which caloric intake is more responsive to income growth (e.g. Carter, Rausser and Smith, 2011). Therefore, Chinese economic activity, and not merely economic activity in the US, appears to drive global food prices.[18]

18 Finally, we have also checked the impulse response of common and sectoral commonalities to Chinese demand shocks at representative points in time: 2003Q4, 2008Q4 and 2013Q4. Consistent with the common perception, we observe a positive response across time for both the common factor and three sectoral factors. The effect is time varying with a peak effect in 2008Q4 – see Figure B1 (Appendix in supplementary data at *ERAE* online).

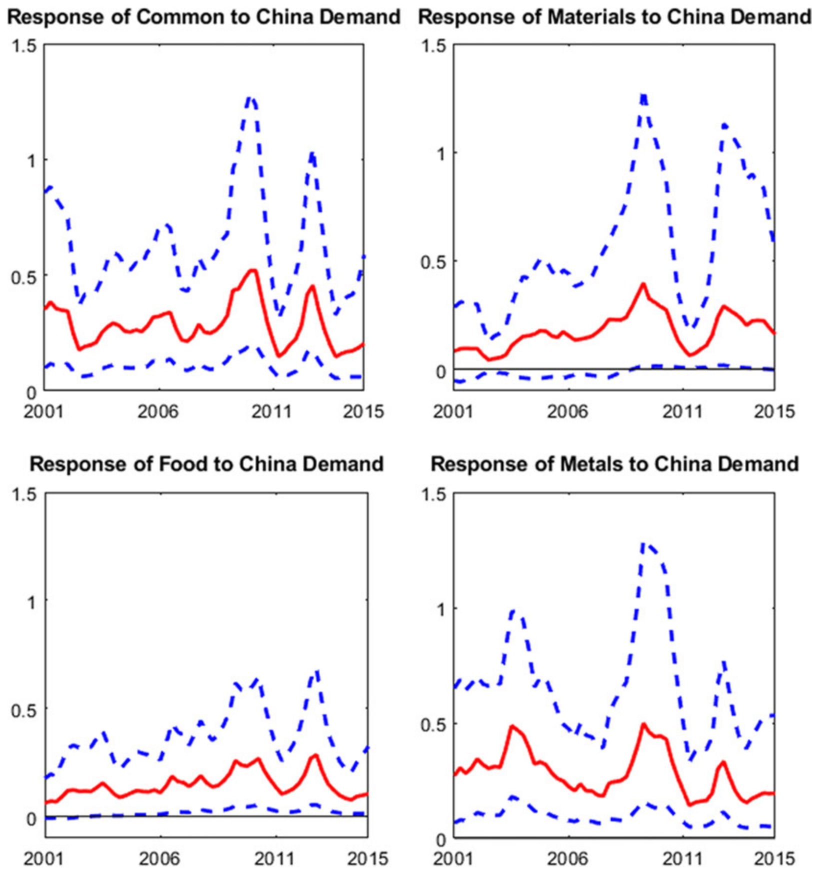

Fig. 5. TVP-FAVAR factor responses to Chinese demand shock. *Notes*: The four graphs in figure plot the median responses in commodity returns of common and three sector factors to Chinese demand shocks, plus 16th and 84th percentile error bands. The estimates are based on the TVP-FAVAR model in equation (11). The sample period is based on data availability.

5.4. Robustness checks

We have carried out several robustness check on our main results and results are in Appendix B (in supplementary data at *ERAE* online). Our first robustness check focuses on providing evidence that our assumptions on the identification order adopted in our impulse response analysis do not affect our main empirical findings.[19] First, we use the following Cholesky ordering: real interest rate (R_t), real economic activity ($Demand_t$), risk ($Risk_t$) and the common factor (F_t). Figures B2 and B3 (in Appendix B in supplementary data at *ERAE* online) show that the directions of the responses to the

19 We would like to thank one of the anonymous referees, who suggested considering the robustness of our identification order.

structural shocks from both constant FAVAR and time-varying parameter FAVAR with stochastic volatility models are qualitatively similar to our main results (see Figures 2 and 3). Note that we also perform the same exercises for sectoral factors ($Sectoral_{b,t}$) and the new set of results in Figure B4 (Appendix B in supplementary data at *ERAE* online) are in line with our baseline model results in Figure 4. Furthermore, Figures B5–B7 (Appendix B in supplementary data at *ERAE* online) report the results with four variables ordered as follows: $Y_t = [Risk_t, Demand_t, R_t, F_t \text{ or } Sectoral_{b,t}]$. Again, we find that the responses of commodity commonalities to fundamental shocks do not qualitatively change our main conclusions.

Second, we re-estimate our models by replacing several variables in our main model with alternative proxies, such as Kilian's (2009) global economic activity index, the federal funds rate, realised volatility of commodities and stock market volatility. We report the results in Figures B8–B11 (Appendix B in supplementary data at *ERAE* online), and we find that the responses of commodity commonalities to fundamental shocks are in line with our main results. In addition, we have also tested the robustness of our TVP-FAVAR results by employing different draw numbers used in the DHFM. Figure B12 (Appendix B in supplementary data at *ERAE* online) shows that the responses of commodity commonalities to fundamental shocks are not different from our main findings.

Furthermore, we also check the responses of commodity prices to three macro shocks at three different points in time using the TVP-FAVAR with stochastic volatility models. Figure B13 (Appendix B in supplementary data at *ERAE* online) reports that the responses of common factor are observed at ten-year intervals, i.e. 1988Q4, 1998Q4 and 2008Q4.[20] Furthermore, we have also checked the responses of both common and sector factors to Chinese demand shocks at different times – see Figure B14 (Appendix B in supplementary data at *ERAE* online). In sum, we find strong evidence of heterogenous responses of commodity prices to macro shocks over time and across sector. These findings underscore the importance of allowing for time variation in studying the effects of macro fundamentals on commodity commonalities.

6. Conclusion

Large swings in commodity prices have brought new momentum to the spirited debate on their commonalities and determinants, and our paper extends this literature. We first contribute to the empirical evidence on the co-movement of real primary commodity prices. Unlike existing studies that extract principal components from a large panel of data, we employ a dynamic hierarchical factor model from Moench, Ng and Potter (2013) to decompose the real price of commodities into common, sectoral, sub-sector and idiosyncratic components. We find significant evidence of co-movement in commodity prices and, importantly, identify a

20 There are some notable differences in these responses in the representative periods for the three shocks. For example, we can clearly see increasing responses over time of commodity prices to demand shock; the increasing impact of the real interest rate shock and, also, the risk impulse responses become highly negative and important in 2008Q4.

common factor and three sectoral factors. Our results highlight the importance of modelling common variations at different levels: for instance, common and sectoral factors may share similar trends, but there is also notable heterogeneity.

Next, we empirically relate commodity price commonalities to (i) demand, (ii) real interest rates and (iii) uncertainty. While existing studies often present conflicting evidence of the impact of fundamentals using time-invariant methodologies to examine the drivers of commodity prices co-movement, this study uses a time-varying parameter factor augmented vector autoregression (TVP-FAVAR) model with stochastic volatility. This allows us to capture potentially unstable relationships to fundamentals and a time-varying impact on the commodity market.

The results from this analysis can be summarised as follows. First, we show that positive innovations to the global business cycle cause a higher price of commodities over time, especially after the 2000s. Second, real interest rate shocks were found to have an important and negative effect on real commodity commonalities. Furthermore, we find that elevated risks negatively impact commodity prices. This uncertainty effect on the real price of commodities was most acute from the middle of 2007 and peaked at the end of 2008. This finding is consistent with the idea that many asset classes suffered from the adverse impact of the 2007–2008 global financial crisis. Last but not the least, we extend the literature on the relationship between commodity prices and macro determinants, while allowing for commodity heterogeneity. We find important heterogeneity in the response of agricultural raw material, food and metal sectors. For example, the material and metal sectors in general respond more positively to the global business cycles and risk, while real interest rate shocks have a more persistent impact on food prices.

The results from this analysis can be summarised as follows. First, we found that positive innovations to the global business cycle cause a higher price of commodities over time, especially after the 2000s. Second, real interest rate shocks were found to have an important and negative effect on real commodity commonalities. Furthermore, we found that elevated risks negatively impact commodity prices. This uncertainty effect on the real price of commodities was most acute from the middle of 2007 and peaked at the end of 2008. This finding is consistent with the idea that many asset classes suffered from the adverse impact of the 2007–2008 global financial crisis. Finally, we extended the literature on the relationship between commodity prices and macro determinants, while allowing for commodity heterogeneity. We found sectoral factors for agricultural raw material, food and metal sectors responded differently to macro shocks at different points in time.

In sum, our findings provide useful information for macroeconomic policy making, consumption, capital investment, risk and portfolio management. Our results suggest that the impact of the uncertainty shock at the global financial crisis is large and it requires the central bank to cut the interest rate aggressively. This finding supports the discussion of Mishkin (2009) who proposes aggressive monetary policies during crises. Our results also imply that the impact of the demand shock can be mitigated by monetary policies.

Moreover, our findings provide useful information for policy makers that the effects of monetary policy vary over time and across sectors. Monetary authorities need to take these relationships into account to ensure they produce their desired policy outcome. Recently, the US Federal Reserve began to raise policy rates, and our empirical results indicate that this will impact commodity prices. Our results are therefore informative for commodity exporting and importing countries alike. They need to consider the prices of imported and exported commodities and the effects of US monetary policy. The spillovers of US monetary policy clearly play an important role in commodity orientated countries (Miranda-Agrippino and Rey, 2018). One interesting question will be to find out whether our time-varying framework can improve the predictability of commodity prices. We shall leave these issues for future work.

Supplementary data

Supplementary data are available at *European Review of Agricultural Economics* online

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